17 There are 9 counters in a bag.

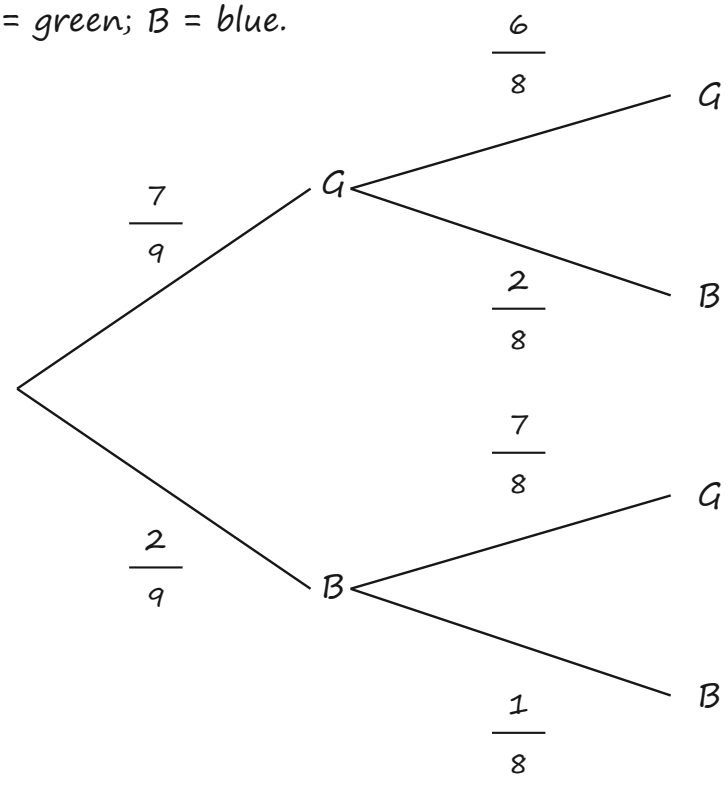
7 of the counters are green.
2 of the counters are blue.

Ria takes at random two counters from the bag.

Work out the probability that Ria takes one counter of each colour.
You must show your working.

No replacement: only 8 counters remain.

G = green; B = blue.



1 mark
Denominators become 8

G then B: $\frac{7}{9} \times \frac{2}{8} = \frac{14}{72}$

B then G: $\frac{2}{9} \times \frac{7}{8} = \frac{14}{72}$

1 mark
Multiply along a path

P(one of each) = $\frac{14}{72} + \frac{14}{72} = \frac{28}{72}$

1 mark
Add both possible orders

Simplify: $\frac{28}{72} = \frac{7}{18}$

1 mark
Correct probability

$\frac{7}{18}$

(Total for Question 17 is 4 marks)